Viva survival sheet

If you only read one page before the viva, read this one.

1. The six commands that answer most questions

Run these in a terminal, or in the Azure portal: **aks-cicd-cia2** → **Run command** (left menu, under Settings). Everything lives in the namespace `cicd-demo`.

```
kubectl -n cicd-demo get deployment,pods,svc      # what is running
kubectl -n cicd-demo describe pod <pod>           # why is it unhappy (read Events)
kubectl -n cicd-demo logs <pod>                   # what the app printed
kubectl -n cicd-demo scale deployment/cicd-demo-app --replicas=3
kubectl -n cicd-demo rollout history deployment/cicd-demo-app
kubectl -n cicd-demo rollout undo deployment/cicd-demo-app
```

If you remember nothing else: `get` shows you what exists, `describe` tells you why it is misbehaving, `logs` shows what the app said. Almost every "look at X" question is one of those three.

2. If she asks you to CHANGE something

She says	You type
Add another pod / scale up	kubectl -n cicd-demo scale deployment/cicd-demo-app --replicas=3
Go back to the previous version	kubectl -n cicd-demo rollout undo deployment/cicd-demo-app
Change a setting the app reads	kubectl -n cicd-demo edit configmap cicd-demo-app-config then: kubectl -n cicd-demo rollout restart deployment/cicd-demo-app
Change anything in the pod spec	kubectl -n cicd-demo edit deployment/cicd-demo-app
Restart the app	kubectl -n cicd-demo rollout restart deployment/cicd-demo-app
Deploy a different image	kubectl -n cicd-demo set image deployment/cicd-demo-app app=acrcicdcia2.azurecr.io/cicd-demo-app:8-95e0d05
Add a node to the cluster	az aks scale -g rg-cicd-cia2 -n aks-cicd-cia2 --node-count 2

Watch what happens afterwards - that is half the marks: `kubectl -n cicd-demo get pods -w` (Ctrl+C to stop).

3. The one idea that explains everything

Kubernetes is a **thermostat, not a script**. You never tell it what to do. You tell it what should be true - "two copies of this image should be running" - and a controller loops forever: read the desired state, look at reality, fix the difference.

That single sentence answers a huge range of questions. Why did the pod come back when I deleted it? The ReplicaSet saw 1 where it wanted 2. Why did my manual change get undone? Something else owns that number. What happens if a node dies? The pods are rescheduled, because the desired state still says they should exist.

spec = what you want. **status** = what you have. The whole system is the loop closing the gap between those two.

4. When you genuinely do not know

Do not freeze and do not guess. Say what you would *do*. This is a real engineering answer and it scores better than a wrong fact:

"I am not certain off the top of my head, but I would find out by running `kubectl describe` on it and reading the Events, because that is where Kubernetes reports why it could not do what was asked."

Then actually run it. Reading output calmly in front of the examiner is a better demonstration of competence than reciting something memorised.

5. Things that LOOK broken but are not

Symptom	What is really happening
Pod name changed after a redeploy	Pods are disposable. Every new version creates new pods with new names. The Service keeps the stable address - that is what it is for
A manual scale got reverted	An HPA owned the replica count. It has been removed from this cluster, so scaling now sticks
Service shows EXTERNAL-IP pending	Azure takes 2-5 minutes to create the load balancer. Wait, do not re-apply
Pod says Pending	It does not fit on the node. describe pod, read Events: "Insufficient cpu". Fix: fewer replicas, or add a node
Pod says ImagePullBackOff	It cannot pull the image. Usually the AcrPull permission or a typo in the image name
Changed the ConfigMap, nothing happened	Environment variables are read once at container start. Needs a rollout restart
Builds 2-5 are red in Jenkins	They ran before SonarCloud was configured. The Quality Gate correctly refused to pass. Builds 8 and 11 are the green ones
14 extra pods in kube-system	Kubernetes' own components - DNS, metrics, networking. AKS installs them; you did not

6. Where everything lives

Thing	Where
The running app	http://cicd-demo-cia2.centralindia.cloudapp.azure.com
Jenkins	http://localhost:8090 (user admin)
Pods in the portal	aks-cicd-cia2 → Kubernetes resources → Workloads → namespace cicd-demo
Run kubectl in the portal	aks-cicd-cia2 → Run command (left menu, under Settings)
Images	acrcicdcia2 → Repositories → cicd-demo-app
Code	github.com/manisudhan/cicd-azure-cia2
Code quality	sonarcloud.io → manisudhan_cicd-azure-cia2
The full explanation	docs/CICD-Study-Guide.pdf in the repo

7. Numbers worth knowing cold

Fact	Value
Pipeline stages	8
Unit tests	29, all passing
Code coverage	95.7% instruction, 88.5% branch
Replicas	2
Image tag format	<build>-<commit>, e.g. 11-ac268ea
Why not 'latest'	kubectl apply would report "unchanged" and nothing would redeploy
Rolling update	maxSurge 1, maxUnavailable 0 - new pod Ready before an old one goes
Node	1 x Standard_B2s, 2 vCPU, 4 GiB
Cost	about 2 USD per day
The two gates	Gitleaks (secrets) and SonarQube Cloud (quality) - both FAIL the build

8. The 30-second demo, if you get to choose

Open `http://cicd-demo-cia2.centralindia.cloudapp.azure.com` in a browser. In a terminal, delete a pod. Refresh the browser - still up. Show the new pod.

```
kubectl -n cicd-demo get pods
kubectl -n cicd-demo delete pod <one-of-them>
kubectl -n cicd-demo get pods      # a replacement already exists
```

This single demo shows self-healing, the Service abstraction and zero downtime - three concepts in half a minute, with no slides. Nobody told Kubernetes to replace that pod; the ReplicaSet noticed the count was wrong and fixed it.